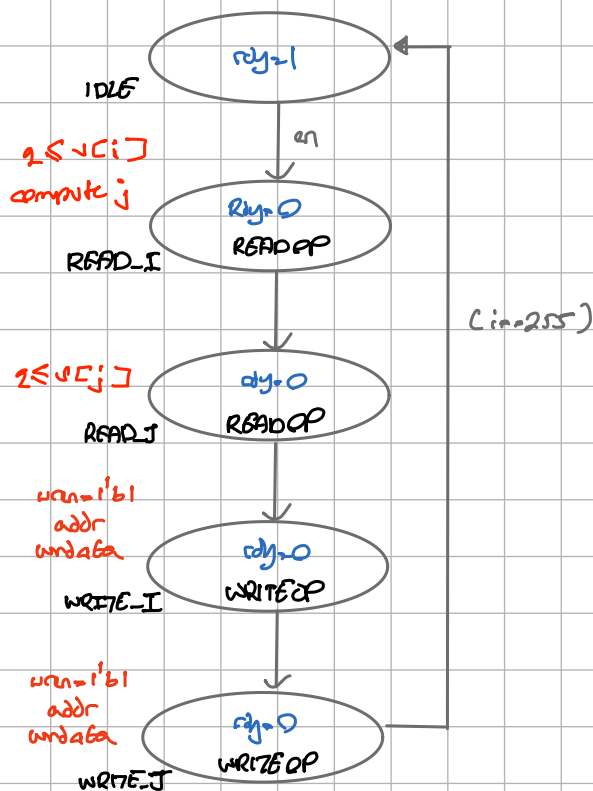


state diagramPseudo code

```

j = 0
for i = 0 to 255:
    j = (j + v[i] +
    key[i mod 3]) mod 256
    swap(v[i], v[j])
  
```

SINGLE PORT READ

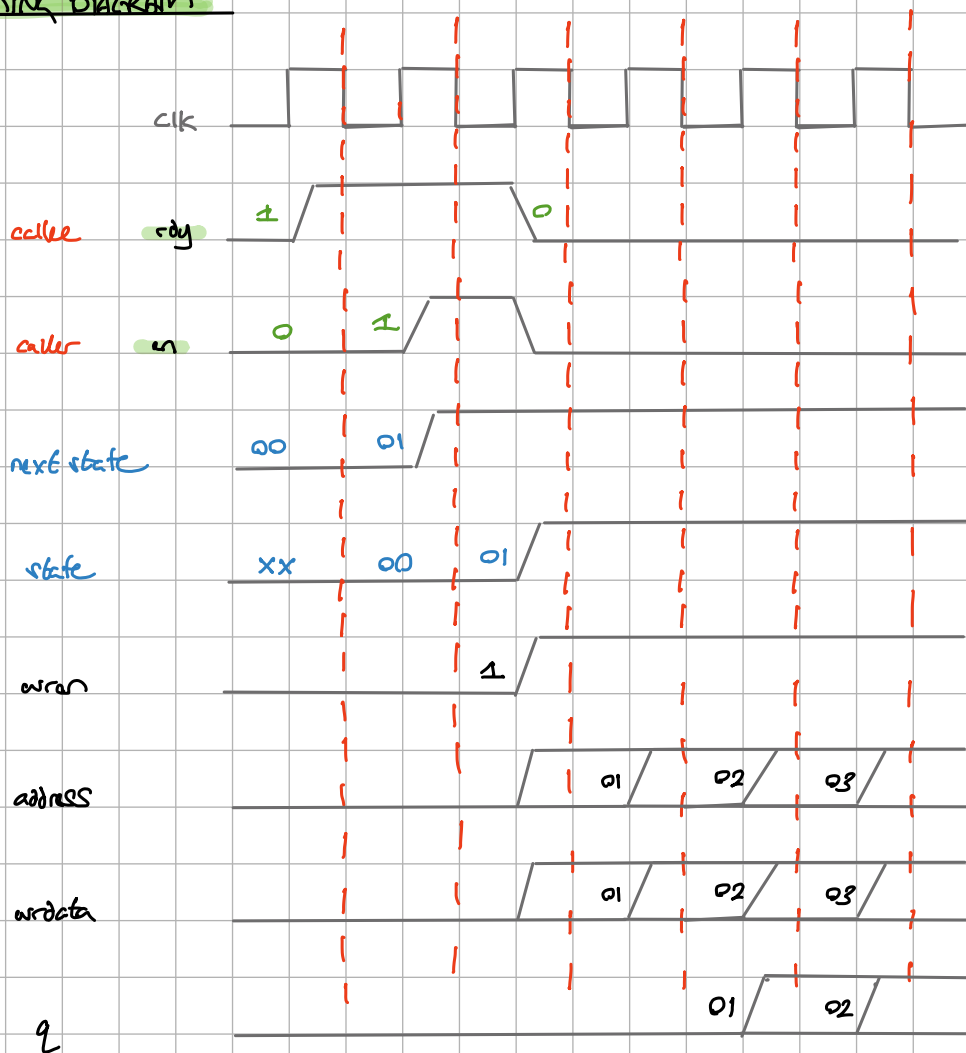
```

addr = add
z = data
wdata = x
wren = x
q <= wmem(addr)
  
```

SINGLE PORT WRITE

```

wren = i'b1;
address = mem.addr
wdata = data
wmem[address] = wdata
  
```

TIMING DIAGRAM

$$S_C \supseteq id_A + g$$